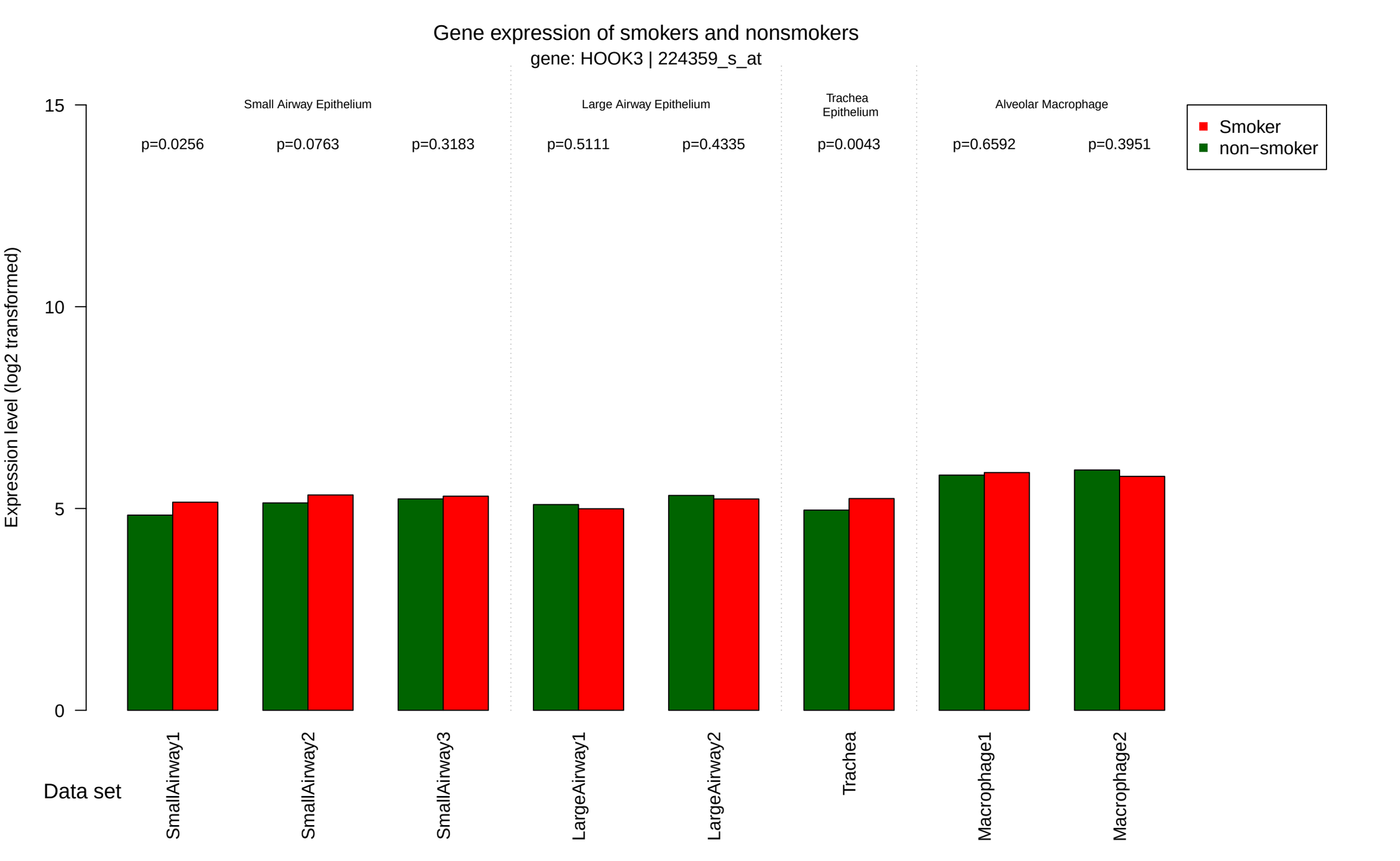
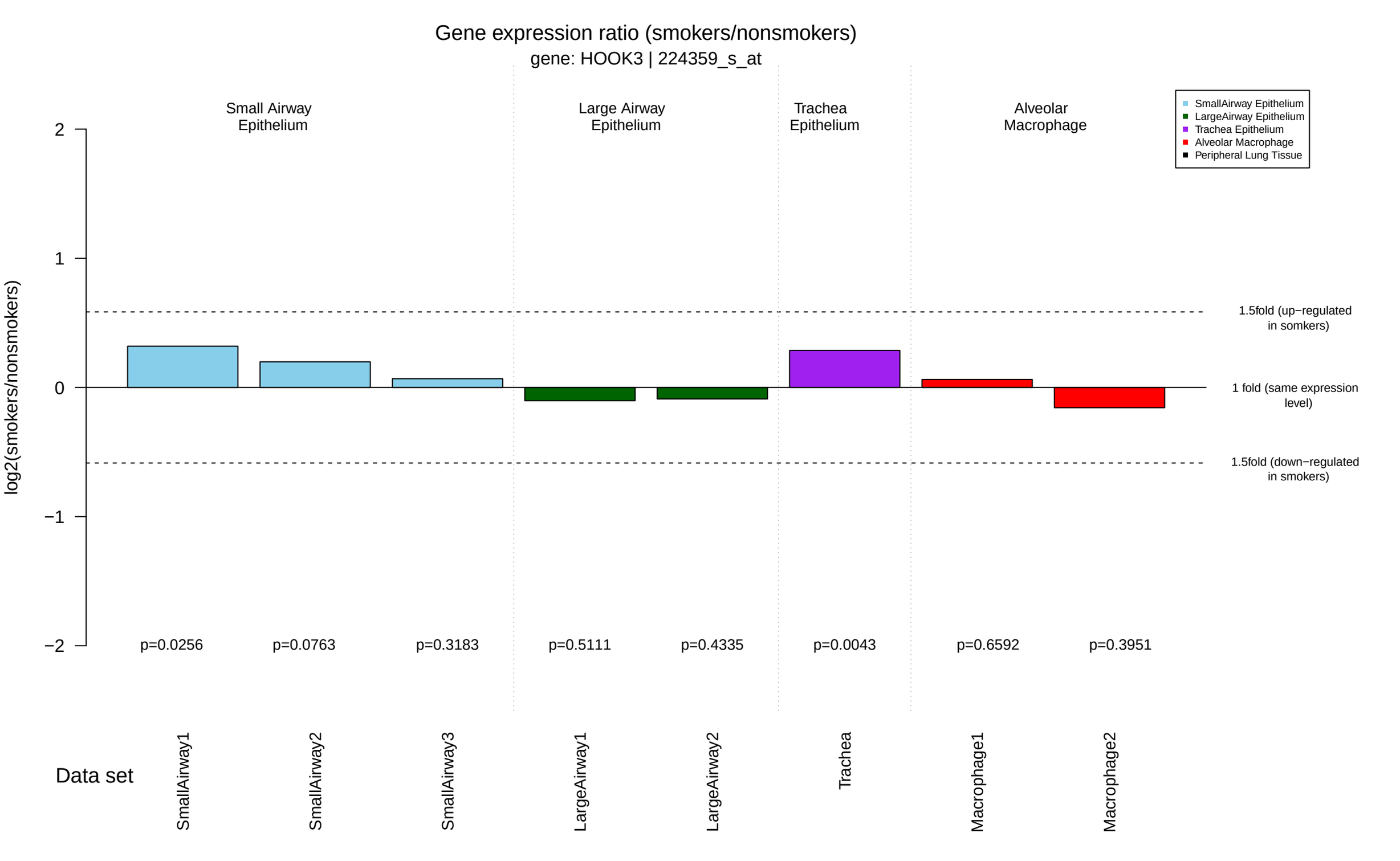
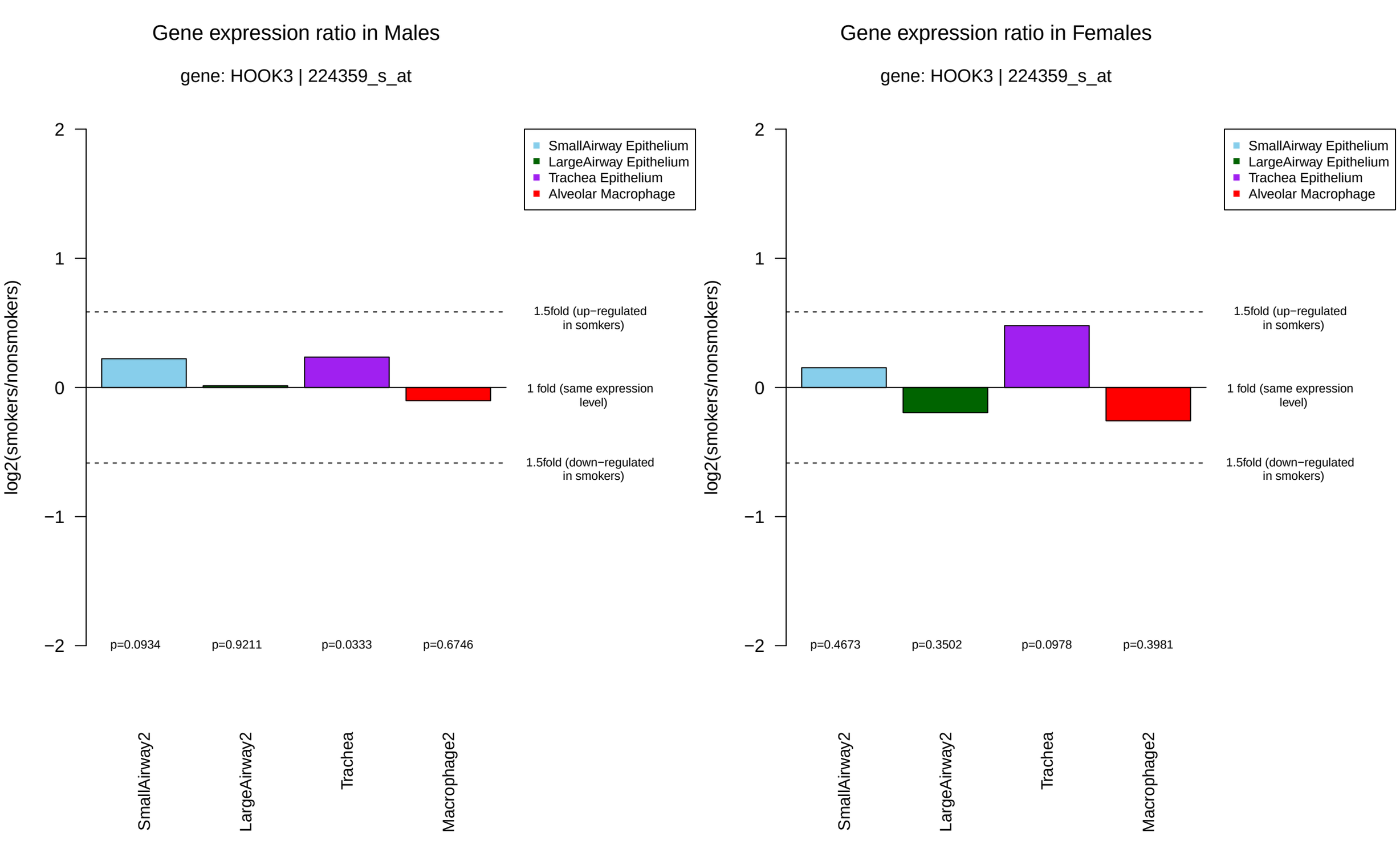
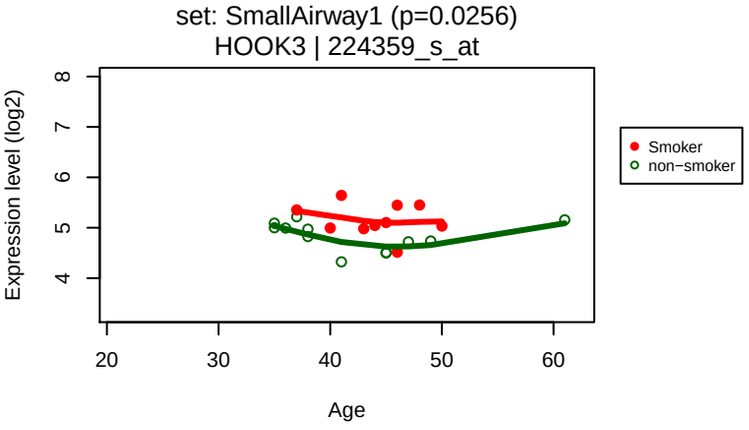
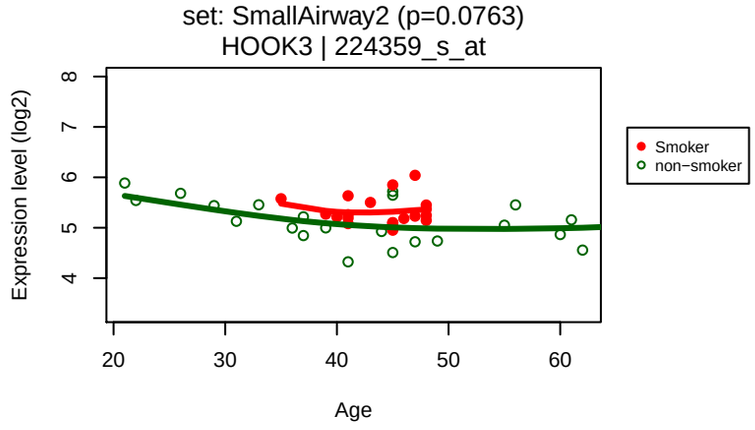
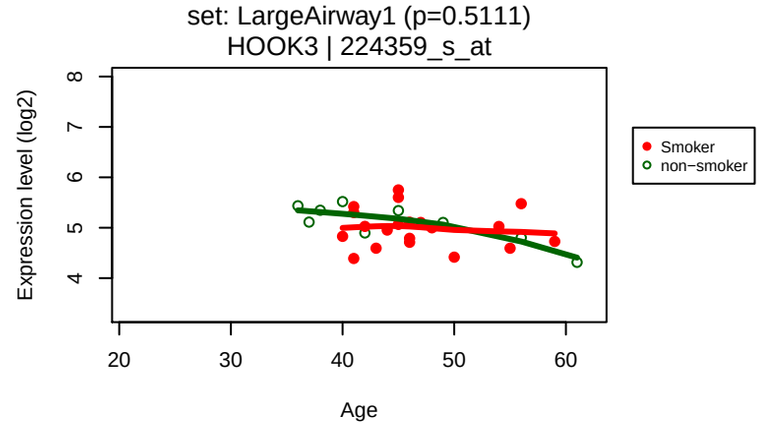

Gene expression of smokers and nonsmokers
gene: HOOK3 | 224359_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0256
p=0.0763
p=0.3183
p=0.5111
p=0.4335
p=0.0043
p=0.6592
p=0.3951
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

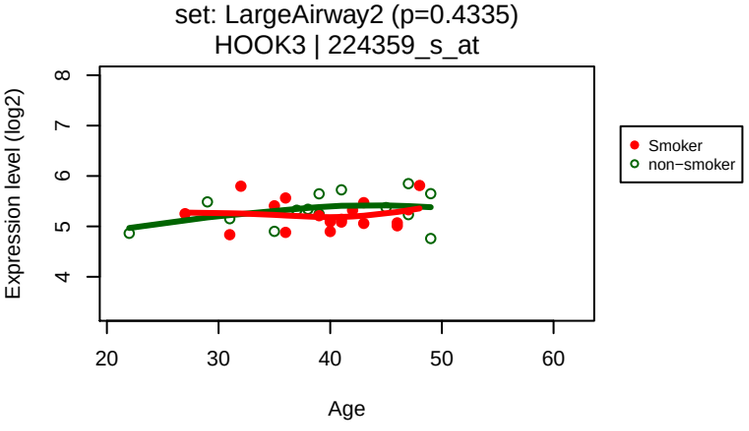

Gene expression ratio (smokers/nonsmokers)
gene: HOOK3 | 224359_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0256
p=0.0763
p=0.3183
p=0.5111
p=0.4335
p=0.0043
p=0.6592
p=0.3951
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

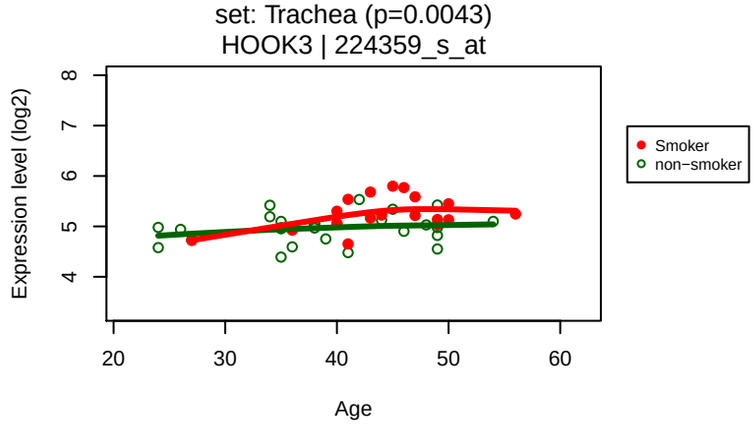

Gene expression ratio in Males
gene: HOOK3 | 224359_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0934
p=0.9211
p=0.0333
p=0.6746
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOOK3 | 224359_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4673
p=0.3502
p=0.0978
p=0.3981
SmallAirway2
LargeAirway2
Trachea
Macrophage2

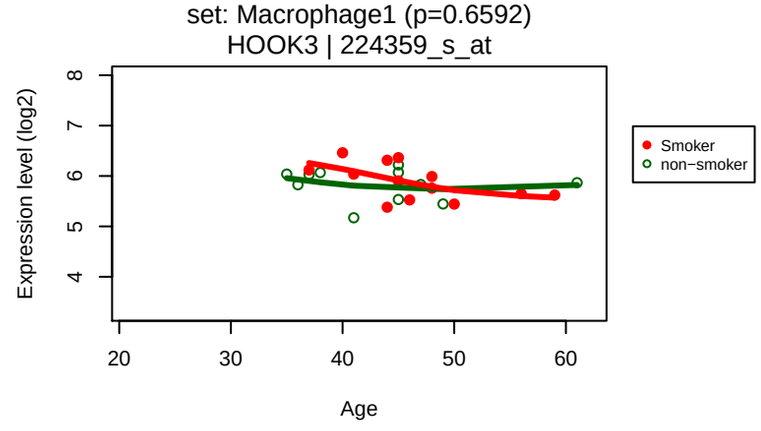

set: SmallAirway1 (p=0.0256)
HOOK3 | 224359_s_at
Expression level (log2)
Age
Smoker
non−smoker

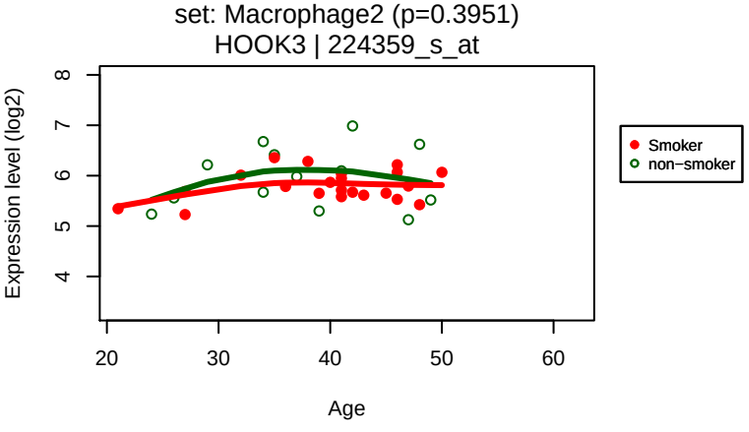

set: SmallAirway2 (p=0.0763)
HOOK3 | 224359_s_at
Expression level (log2)
Age
Smoker
non−smoker

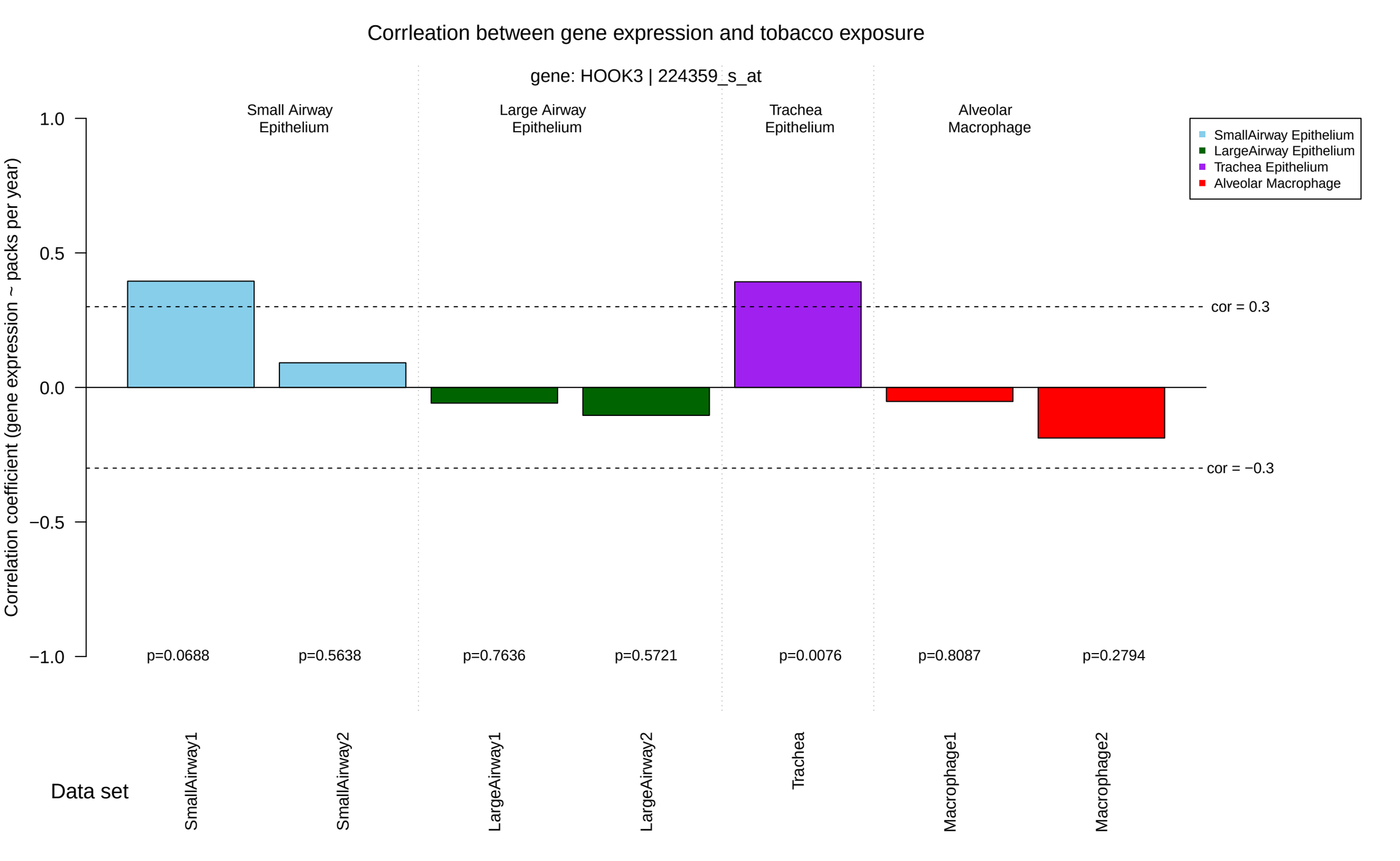

set: LargeAirway1 (p=0.5111)
HOOK3 | 224359_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4335)
HOOK3 | 224359_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0043)
HOOK3 | 224359_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6592)
HOOK3 | 224359_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3951)
HOOK3 | 224359_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOOK3 | 224359_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0688
p=0.5638
p=0.7636
p=0.5721
p=0.0076
p=0.8087
p=0.2794
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2